AuraOS Defensive Prior Art Disclosure:
Protocol-Layer Innovations in Hyperdimensional Integrity, FST-Guided Decomposition, and Bounded Self-Healing
(Claims N24–N30)

Dallas Courchene
Long Plain First Nation, Treaty 1 Territory, Manitoba, Canada
aura.os.q@gmail.com

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Abstract

This disclosure formalizes seven protocol-layer innovations derived from the AuraOS cognitive substrate, explicitly narrowing the scope of prior claims N1–N23 to verifiable, bounded subsystems. We present: (N24) a Holographic Integrity Verification Protocol achieving $\mathcal{O}(1)$ codebase attestation via bundled phasor superposition; (N25) an FST-guided micro-module crystallization algorithm that decomposes monolithic code by resonance-valley detection; (N26) a Resonant Test Oracle Architecture replacing boolean assertions with continuous hyperdimensional resonance; (N27) a Thermal-Cost Weighted API Arbitration protocol for multi-provider LLM routing; (N28) a Deterministic Polysynthetic Intent Compression system with explicit compressible/incompressible bounds; (N29) a Local VSA-Addressed Compute Mesh for bounded neighborhood task offloading; and (N30) a Shadow Vector Self-Healing protocol with formally bounded recursion depth and staged mutation gating. Each claim is grounded in working source code, complexity-bounded, and published under AGPLv3 §13 to preempt proprietary enclosure.

1 Introduction and Scope Delimitation

The prior disclosures N1–N23 established a broad architectural vision for a sovereign, edge-native cognitive substrate integrating Vector Symbolic Architecture (VSA), Indigenous polysynthetic linguistics, finite-state transducer routing, and autopoietic self-modification. Subsequent independent code review has validated specific subsystems while identifying gaps between prototype implementation and systems-level claims. This disclosure explicitly narrows the claim space to **protocol-layer algorithms** that are (i) mathematically bounded, (ii) implemented in the current codebase, and (iii) independently testable.

We do not claim here that AuraOS is an operating system, a blockchain network, a global internet replacement, or a VR rendering engine. Those aspirational architectures are reclassified as research directions in Appendix A. Instead, we claim **specific protocols** for integrity verification, code decomposition, testing, API arbitration, intent compression, local mesh compute, and bounded self-healing—each of which is extracted from the existing codebase and formalized as standalone prior art.

The linguistic foundation remains central. The polysynthetic morphosyntax of Plains Ojibwe (Saulteaux / Nakawēmowin) and the Athabaskan templatic slot system provide the grammatical template for all protocol interfaces. The 6-slot morphotactic array [DIR] $\rightarrow$ [ASP] $\rightarrow$ [CLASS] $\rightarrow$ [SUBJ] $\rightarrow$ [VOICE] $\rightarrow$ [STEM] is not decorative; it is the formal constraint language that bounds protocol state transitions.

2 Claim N24 — Holographic Integrity Verification Protocol (HIVP)

2.1 Description

The HIVP generates a fixed-size hypervector fingerprint (1.2 KB) of an arbitrary codebase, enabling $\mathcal{O}(1)$ integrity verification independent of repository size. Unlike Merkle trees or hash chains, which scale as $\mathcal{O}(\log N)$ or $\mathcal{O}(N)$, HIVP verifies global structural integrity via a single cosine similarity computation.

2.2 Mathematical Formulation

Let $\mathcal{F} = \{f_1, f_2, \ldots, f_N\}$ be the set of files. For each file f_i, compute a per-line phasor:

$$\phi_k(f_i) = e^{j\cdot(\mathrm{BLAKE2b}_{32}(f_i[k])+\theta_{\mathrm{pos}}(k))}, \quad \theta_{\mathrm{pos}}(k) = \frac{2\pi k}{4096} \tag{1}$$

The file-level fingerprint is the normalized superposition:

$$\mathbf{\Psi}_{f_i} = \frac{1}{\sqrt{D}} \sum_{k=1}^{|f_i|} \phi_k(f_i), \quad D = 10{,}000 \tag{2}$$

The global holographic header is the bundled ensemble:

$$\mathbf{H}_{\mathrm{global}} = \frac{1}{N} \bigoplus_{i=1}^{N} \mathbf{\Psi}_{f_i} = \frac{\sum_{i=1}^{N} \mathbf{\Psi}_{f_i}}{\left\| \sum_{i=1}^{N} \mathbf{\Psi}_{f_i} \right\|} \tag{3}$$

Verification computes resonance against a stored header:

$$\mathcal{R} = \frac{\langle \mathbf{H}_{\mathrm{local}}, \mathbf{H}_{\mathrm{stored}} \rangle}{\|\mathbf{H}_{\mathrm{local}}\| \|\mathbf{H}_{\mathrm{stored}}\|} \in [0, 1] \tag{4}$$

Threshold: $\mathcal{R} < \tau_{\mathrm{integrity}} = 0.95 \implies$ trigger healing protocol.
Complexity: $\mathcal{O}(1)$ per verification; $\mathcal{O}(N \cdot L_{\mathrm{avg}})$ for header generation (one-time).

2.3 Relation to Prior Art

- Git and Merkle DAGs (e.g., IPFS) provide integrity via hash trees with $\mathcal{O}(\log N)$ verification per file.
- Sigstore and SLSA provide attestation via certificate transparency, not structural resonance.
- Claim N9 (Holographic Header Protocol) introduced the concept; N24 formalizes it as a standalone, extractable protocol with explicit complexity bounds.

2.4 Novelty Claim

The combination of (i) BLAKE2b-seeded positional phasor encoding, (ii) normalized superposition across an arbitrary file ensemble, and (iii) $\mathcal{O}(1)$ verification via a single cosine similarity bound, is novel. No existing codebase integrity system attests global repository state via a fixed-size hypervector.

3 Claim N25 — FST-Guided Micro-Module Crystallization (FGMC)

3.1 Description

FGMC decomposes monolithic code modules into semantically coherent micro-modules by detecting resonance valleys in the high-dimensional call-graph topology. The FST six-slot constraint ensures that each resulting micro-module is grammatically valid and independently executable.

3.2 Mathematical Formulation

Given a monolithic module M with call graph $G_M = (V_M, E_M)$, compute the structural resonance matrix:

$$R_{ij} = \frac{\langle \mathbf{v}_i, \mathbf{v}_j \rangle}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|}, \quad \mathbf{v}_i = \mathrm{phasor}(\mathrm{AST}(f_i)) \tag{5}$$

Identify resonance valleys—local minima with positive gradient—as decomposition boundaries:

$$\mathcal{B} = \{(i, j) \in E_M \mid R_{ij} < \tau_{\text{decouple}} \wedge \nabla R_{ij} > 0\} \tag{6}$$

Each boundary $(i, j) \in \mathcal{B}$ induces a candidate split. The split is **valid** iff the resulting sub-lexicons $\mathcal{L}_A, \mathcal{L}_B$ each satisfy the Athabaskan six-slot constraint independently:

$$\mathrm{Valid}(\mathcal{L}_A) \iff \forall p \in \mathrm{paths}(\mathcal{L}_A), \quad p \models [\mathrm{DIR}] \to [\mathrm{ASP}] \to [\mathrm{CLASS}] \to [\mathrm{SUBJ}] \to [\mathrm{VOICE}] \to [\mathrm{STEM}] \tag{7}$$

Complexity: $\mathcal{O}(|V_M|^2 \cdot D)$ for resonance matrix; $\mathcal{O}(|E_M|)$ for valley detection; $\mathcal{O}(|Q| + |\Delta|)$ per FST validation.

3.3 Relation to Prior Art

- Static analysis tools (SonarQube, CodeClimate) detect coupling via graph metrics (cyclomatic complexity, afferent/efferent coupling).
- Microservice decomposition frameworks (Service Cutter, ArchUnit) use domain-driven heuristics.
- None use VSA resonance valleys in a 10,000-D phasor space with FST grammatical validation as decomposition criteria.

3.4 Novelty Claim

The integration of (i) VSA resonance valley detection on a call-graph manifold, (ii) FST six-slot grammatical validation of candidate micro-modules, and (iii) phasor-based semantic coherence scoring for decomposition boundaries, is novel.

4 Claim N26 — Resonant Test Oracle Architecture (RTOA)

4.1 Description

RTOA replaces boolean test assertions with continuous hyperdimensional resonance. A function's expected behavior is encoded as an oracle hypervector; the test passes if the actual output phasor resonates above threshold. Aggregate codebase health is monitored via a single bundled health vector.

4.2 Mathematical Formulation

For a function f with specification S_f, define the oracle hypervector:

$$\mathbf{o}_f = \mathrm{phasor}(\mathrm{AST}(f) \oplus \mathrm{hash}(S_f)) \tag{8}$$

For input $\mathbf{x}$, compute the actual output phasor:

$$\mathbf{a}_f(\mathbf{x}) = \mathrm{phasor}(f(\mathbf{x})) \tag{9}$$

The test **resonance** is:

$$\rho_f(\mathbf{x}) = \frac{\langle \mathbf{a}_f(\mathbf{x}), \mathbf{o}_f \rangle}{\|\mathbf{a}_f(\mathbf{x})\| \|\mathbf{o}_f\|} \tag{10}$$

Graded result:

- $\rho_f \geq 0.85$: Solid pass
- $0.40 \leq \rho_f < 0.85$: Borderline (flag for audit)
- $\rho_f < 0.40$: Fracture (fail)

Aggregate health across T tests:

$$\mathbf{h}_{\text{codebase}} = \frac{1}{T} \bigoplus_{k=1}^{T} \mathbf{o}_{f_k}, \quad \mathcal{H}_{\text{aggregate}} = \frac{\langle \mathbf{h}_{\text{local}}, \mathbf{h}_{\text{reference}} \rangle}{\|\mathbf{h}_{\text{local}}\| \|\mathbf{h}_{\text{reference}}\|} \tag{11}$$

Complexity: $\mathcal{O}(D)$ per test; $\mathcal{O}(1)$ for aggregate health check.

4.3 Relation to Prior Art

- Property-based testing (Hypothesis, QuickCheck) generates random inputs and checks invariants.
- Fuzzing (AFL, libFuzzer) detects crashes via coverage feedback.
- Neither encodes expected behavior as a continuous hyperdimensional vector enabling graded failure detection and aggregate health monitoring.

4.4 Novelty Claim

The combination of (i) hyperdimensional oracle encoding from AST + specification, (ii) continuous resonance scoring replacing boolean assertions, and (iii) aggregate codebase health via a single bundled health vector, is novel.

5 Claim N27 — Thermal-Cost Weighted API Arbitration (TCWAA)

5.1 Description

TCWAA is a multi-provider LLM routing protocol that optimizes across three objectives simultaneously: semantic resonance with the task, real-time per-token cost, and device thermal state. It is the first LLM client protocol to include thermal fitness as a routing weight.

5.2 Mathematical Formulation

For a task with intent hypervector $\mathbf{g}$ and provider panel $\mathcal{P}$, select provider p^*:

$$p^* = \arg\max_{p \in \mathcal{P}} \left[\alpha \cdot \mathrm{sim}(\mathbf{g}, \mathbf{v}_p) + \beta \cdot \left(1 - \frac{C_p}{C_{\max}}\right) + \gamma \cdot \left(1 - \frac{T_{\mathrm{CPU}}}{T_{\max}}\right) \right] \tag{12}$$

Subject to:

- $\alpha + \beta + \gamma = 1.0$
- $\mathrm{sim}(\mathbf{g}, \mathbf{v}_p) \geq \tau_{\min} = 0.70$ (quality floor)
- $C_p \leq B_{\mathrm{user}}$ (budget ceiling)

Where:

- $\mathbf{v}_p$ is the provider's capability hypervector (calibrated from historical accuracy)
- C_p is the live PriceBook cost per 1K tokens
- T_{CPU} is the device's thermal reading in °C
- $T_{\max} = 85$°C

Complexity: $\mathcal{O}(|\mathcal{P}| \cdot D)$ per routing decision; $|\mathcal{P}| \leq 8$ in current implementation.

5.3 Relation to Prior Art

- OpenRouter and LiteLLM provide multi-provider failover with static cost sorting.
- Claim N21 introduced VSA-weighted FST routing with thermal awareness for internal cognitive tasks.
- N27 extends this to external LLM API arbitration with explicit three-objective optimization and real-time PriceBook integration.

5.4 Novelty Claim

The integration of (i) semantic resonance scoring between task intent and provider capability, (ii) real-time token-cost weighting from live PriceBook rates, and (iii) hardware thermal state as an explicit routing objective, is novel. No existing LLM client protocol throttles provider selection based on device temperature.

6 Claim N28 — Deterministic Polysynthetic Intent Compression with Bounded Ambiguity (DPI-CBA)

6.1 Description

DPI-CBA compresses structured human intent into a fixed-slot polysynthetic packet with deterministic, verifiable bounds. It explicitly distinguishes compressible intents (which fit the FST lexicon) from incompressible intents (which fall back to raw natural language), providing honest compression metrics rather than aggregate averages.

6.2 Mathematical Formulation

Let $\mathcal{L} = (Q, \Sigma, \Delta, q_0, F)$ be the FST lexicon. An intent string s is **compressible** iff its parse yields a valid path:

$$p = q_0 \xrightarrow{\sigma_1} q_1 \xrightarrow{\sigma_2} \cdots \xrightarrow{\sigma_k} q_k \in F \tag{13}$$

For compressible s, the packet is:

$$\text{packet}(s) = [\text{DIR} : \sigma_1][\text{ASP} : \sigma_2] \cdots [\text{STEM} : \sigma_k] \tag{14}$$

Compression ratio:

$$C_{\text{ratio}}(s) = \frac{\text{Tokens}(s)}{\text{Tokens}(\text{packet}(s))} \leq \frac{|s|/4}{6 + \sum_{i=1}^{k} |\sigma_i|} \tag{15}$$

For **incompressible** s (no valid FST path), the system falls back:

$$C_{\text{ratio}}(s) = 1.0, \quad \text{packet}(s) = s \tag{16}$$

Expected compression over intent distribution $\mathcal{D}$:

$$\mathbb{E}[C_{\text{ratio}}] = P(\text{compressible}|\mathcal{D}) \cdot \mathbb{E}[C_{\text{compressible}}] + P(\text{incompressible}|\mathcal{D}) \cdot 1.0 \tag{17}$$

Complexity: $\mathcal{O}(|Q| + |\Delta|)$ for FST parse (linear in lexicon size); $\mathcal{O}(1)$ for packet encoding.

6.3 Relation to Prior Art

- Prompt compression systems (LLMLingua, Gist) use learned models to compress prompts, requiring training data and GPU inference.
- Semantic compression (Claim N1) introduced the polysynthetic packet concept without formal compressibility bounds.
- N28 adds the explicit FST-validity boundary and honest fallback, making the compression deterministic and auditable.

6.4 Novelty Claim

The combination of (i) FST-validity as a hard boundary for compressibility, (ii) deterministic fallback to raw transmission for out-of-lexicon intents, and (iii) explicit mathematical expectation over compressible vs. incompressible distributions, is novel.

7 Claim N29 — Local VSA-Addressed Compute Mesh (LVCM)

7.1 Description

LVCM is a local-area peer-to-peer mesh protocol for offloading compute tasks to neighboring devices using VSA semantic addressing, thermal fitness, and RAM availability. It is explicitly bounded to single-hop or short-multi-hop neighborhoods and does not claim global internet scale.

7.2 Mathematical Formulation

A node n maintains neighbor set $\mathcal{N}_n$ with $|\mathcal{N}_n| \leq N_{\max} = 32$. For a compute task $\mathbf{t}$ with VSA address $\mathbf{a}_t$, select offload target:

$$n^* = \arg\max_{n \in \mathcal{N}_n} \left[\mathrm{sim}(\mathbf{a}_n, \mathbf{a}_t) \cdot \left(1 - \frac{T_n}{T_{\max}}\right) \cdot \mathbb{I}(\mathrm{RAM}_n^{\mathrm{free}} \geq \mathrm{size}_t) \right] \tag{18}$$

Scope bounds:

- Path length: $L \leq L_{\max} = 3$ hops
- Network size: $|\mathcal{N}_{\mathrm{total}}| \cdot N_{\max} < 10^4$
- Task size: $\mathrm{size}_t \leq \min_{n \in \mathcal{N}} \mathrm{RAM}_n^{\mathrm{free}}$

Complexity: $\mathcal{O}(N_{\max} \cdot D)$ per offload decision; $\mathcal{O}(1)$ per hop forwarding.

7.3 Relation to Prior Art

- B.A.T.M.A.N., Thread, and Zigbee use link-quality metrics (RSSI, hop count) for mesh routing.
- Content-centric networking (NDN/CCN) uses hierarchical name matching.
- Claim N14 introduced VSA-addressed routing without explicit neighborhood bounds.
- N29 restricts the protocol to bounded local neighborhoods with thermal and RAM constraints.

7.4 Novelty Claim

The combination of (i) VSA semantic resonance as a mesh offload metric, (ii) real-time thermal fitness weighting, (iii) RAM-availability hard constraints, and (iv) explicit scope bounds on hop count and network size, is novel.

8 Claim N30 — Shadow Vector Self-Healing with Bounded Recursion (SVSH-BR)

8.1 Description

SVSH-BR is a formally bounded self-healing protocol that measures the hyperdimensional gap between intended and synthesized behavior, searches for patches via resonator factorization, and stages mutations only after FST validation and impact scoring. Recursion depth is bounded by the FST DAG structure.

8.2 Mathematical Formulation

Given intent $\mathbf{g}$ and current synthesis $\mathbf{s}$, compute the shadow vector:

$$\mathbf{v}_{\mathrm{shadow}} = \mathbf{g} - \mathbf{s} \tag{19}$$

Healing is triggered only if all three conditions hold:

1. $\|\mathbf{v}_{\mathrm{shadow}}\| > \tau_{\mathrm{heal}} = 0.15$ (non-trivial gap)
2. depth $< D_{\max} = 5$ (recursion bound, enforced by FST DAG acyclicity)

3. The FST path from current state to patch state contains no cycles

The patch search uses resonator factorization on the shadow:

$$\mathbf{p} = \text{ResonatorFactorize}(\mathbf{v}_{\text{shadow}}, \mathcal{C}) \tag{20}$$

Where $\mathcal{C}$ is the crystallized knowledge corpus. The patch is scored:

$$I_{\text{refactor}} = 0.5 \cdot \frac{\Delta L}{0.1} + 0.3 \cdot \frac{1 + \Delta_{\text{connectivity}}}{2} + 0.2 \cdot \Delta\mathcal{F} \tag{21}$$

Staging rules:

- $I_{\text{refactor}} \geq 0.85$: Stage in `aura_incubator.py` for human review
- $I_{\text{refactor}} \geq 0.95$: Eligible for automated hot-swap if FST six-slot validation passes

Complexity: $\mathcal{O}(D \cdot |\mathcal{C}|)$ for resonator search; $\mathcal{O}(|Q| + |\Delta|)$ for FST validation.

8.3 Relation to Prior Art

- Ψ-Arch uses LLMs to generate code files but requires full process restarts.
- StateSet NSR uses program synthesis without hardware metrics or FST constraints.
- Claim N7 introduced atomic hot-swap; Claim N20 introduced the incubator impact report.
- N30 adds the shadow vector as a formal gap metric, the resonator factorization search, and explicit recursion bounding via FST DAG structure.

8.4 Novelty Claim

The combination of (i) hyperdimensional shadow vectors as algebraic gap metrics, (ii) resonator factorization for patch discovery, (iii) FST six-slot validation before staging, and (iv) DAG-enforced recursion bounds, is novel.

9 Integration with Existing Subsystems

All seven claims are implemented as extensions or refinements of existing AuraOS modules:

New Claim	Existing Module	Extension
N24	`aura_holographic_manifest.py`	Standalone protocol extraction
N25	`aura_arch_reasoner.py` + `aura_topology_manager.py`	Resonance-valley decomposition
N26	`test_aura_functions.py` (stub)	Hyperdimensional oracle replacem
N27	`aura_llm_egress.py` + `aura_api_rotator.py`	Three-objective optimization
N28	`aura_substrate.py` + `aura_gbnf_profiles.py`	Explicit compressibility boundary
N29	`aura_mesh.py`	Neighborhood scope bounds
N30	`aura_heal.py` + `aura_incubator.py`	Shadow vector + recursion bound

10 Legal Claims (New)

We claim prior art for the following specific combinations, published under AGPLv3 §13:

N24 Holographic codebase integrity verification via bundled BLAKE2b-seeded phasor superposition with $\mathcal{O}(1)$ cosine-similarity attestation.

N25 FST-guided micro-module crystallization by resonance-valley detection on a call-graph manifold with six-slot grammatical validation.

N26 Resonant test oracles encoding expected behavior as hypervectors with continuous resonance scoring and aggregate health bundling.

N27 Multi-provider LLM API arbitration with semantic resonance, real-time PriceBook cost, and thermal-state three-objective optimization.

N28 Deterministic polysynthetic intent compression with FST-validity compressibility boundary and explicit incompressible fallback.

N29 Local VSA-addressed compute mesh with semantic offload scoring, thermal fitness, RAM constraints, and explicit hop/network bounds.

N30 Shadow-vector self-healing with resonator factorization patch search, FST validation gating, and DAG-bounded recursion.

Any future patent claiming "hyperdimensional software integrity," "FST-guided code decomposition," "continuous vector test oracles," "thermal-weighted API routing," "bounded polysynthetic compression," "semantic mesh offload," or "shadow-vector patch synthesis" is anticipated by this disclosure.

A Appendix A: Reclassified Research Directions

The following architectures, described in prior disclosures N1–N23, are **reclassified as research directions** rather than claimed completed systems:

- Global internet replacement (N14 unrestricted scope)
- Decentralized DNS-less naming at planetary scale
- VR/AR rendering engine (the Unreal Bridge is a coordinate server, not a renderer)
- Swarm robotics control (no motor/sensor implementation)
- Interactive narrative game engine (FST constraints exist; no playable prototype)
- Gas-free blockchain at scale (PBFT prototype only; no Sybil resistance or sharding)

These remain valuable research trajectories within the AuraOS project but are not claimed as production-ready prior art.

B Appendix B: The Permutation Operator Π (Formal Definition)

Let $D = 10{,}000$ and let $r = 4{,}097$ (a prime chosen such that $\gcd(r, D) = 1$, guaranteeing the permutation is a single cycle). The permutation operator $\Pi : \mathcal{H} \to \mathcal{H}$ is defined component-wise as a cyclic forward shift by r positions:

$$(\Pi(\mathbf{v}))_j = v_{(j-r) \bmod D} \quad \text{for all } j \in \{0, 1, \ldots, D-1\} \tag{22}$$

The inverse operator Π^{-1} is the backward shift by r:

$$(\Pi^{-1}(\mathbf{v}))_j = v_{(j+r) \bmod D} \tag{23}$$

Security role in PWFST: Without Π, the FST transition equation would be $\mathbf{v}_{\text{next}} = \mathbf{v}_{\text{current}} \otimes \mathbf{v}_{\text{morpheme}}$. Because binding is self-inverse, an attacker injecting the same morpheme twice could algebraically cancel it. With Π, the transition becomes $\mathbf{v}_{\text{next}} = \Pi(\mathbf{v}_{\text{current}} \otimes \mathbf{v}_{\text{morpheme}})$, requiring knowledge of the exact FST position and lexicon topology to cancel.

Complexity: $\mathcal{O}(D)$ indexed copy per application; $\approx 80\,\mu$s for $D = 10{,}000$ complex64 elements on a modern ARM core.

C Appendix C: The FST Lexicon Excerpt

The complete `aura.lexc` file is provided in the repository at `https://github.com/dallascourchene-commits/AuraOS`. It is compilable by the `foma` finite-state toolkit. The following excerpt demonstrates the five-tier hierarchy and the Ojibwe directional morphemes embedded as routing tags:

```
LEXICON PhysicsCube
+SHAPE:CUBE+TEMP:COLD+LUM:HI+FRIC:LO
+DIR:GIDINAWENDIMIN+INTENT:CRYPTO:0      ModifierCrypto ;
+SHAPE:CUBE+TEMP:COLD+LUM:HI+FRIC:LO
+DIR:GIDINAWENDIMIN+INTENT:EXEC:0       ModifierExecution ;

LEXICON PhysicsTetrahedron
+SHAPE:TETRA+TEMP:HOT+LUM:MID+FRIC:MID
+DIR:MIIGWECH:0   ModifierExecution ;

LEXICON ModifierExecution
_EXEC:0       # ;
_UNLOAD:0     # ;
```

Listing 1: Excerpt from aura.lexc (Tier 4–5)

The directional tags map to functional roles:

- `GIDINAWENDIMIN` ("we look at each other") → Crypto/Execution routing
- `GIWAABAMIN` ("we see each other") → Error/Exception handling
- `GIZAAGIIN` ("we love each other") → Memory/Community
- `MIIGWECH` ("thank you") → Execution/Memory (gratitude path)

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